

21. Let the number be x .

$$\begin{aligned}\text{Then, } \frac{2}{3}x - 50 &= \frac{1}{4}x + 40 \Leftrightarrow \frac{2}{3}x - \frac{1}{4}x = 90 \\ \Leftrightarrow \frac{5x}{12} &= 90 \Leftrightarrow x = \left(\frac{90 \times 12}{5}\right) = 216.\end{aligned}$$

22. Let the number be x .

$$\begin{aligned}\text{Then, } \left[\frac{(x/2)}{6} + \frac{(x/2)}{4}\right] - \frac{x}{5} &= 4 \Leftrightarrow \frac{x}{12} + \frac{x}{8} - \frac{x}{5} = 4 \\ \Leftrightarrow \frac{10x + 15x - 24x}{120} &= 4 \\ \Leftrightarrow x &= 4 \times 120 = 480.\end{aligned}$$

23. Let the number be x .

$$\text{Then, } \frac{x}{3} - \frac{(x+1)}{4} = 1 \Leftrightarrow 4x - 3(x+1) = 12 \Leftrightarrow x = 15.$$

24. Let the number be x .

$$\begin{aligned}\text{Then, } x + x^2 &= 182 \Leftrightarrow x^2 + x - 182 = 0 \\ \Leftrightarrow (x + 14)(x - 13) &= 0 \Leftrightarrow x = 13.\end{aligned}$$

25. Let the number be x .

$$\begin{aligned}\text{Then, } x^2 - (73)^2 &= 5075 \Leftrightarrow x^2 - 5329 = 5075 \\ \Leftrightarrow x^2 &= 5075 + 5329 = 10404 \\ \Leftrightarrow x &= \sqrt{10404} = 102.\end{aligned}$$

26. Let the integer be x .

$$\begin{aligned}\text{Then, } x^2 - 20x &= 96 \\ \Leftrightarrow x^2 - 20x - 96 &= 0 \\ \Leftrightarrow (x + 4)(x - 24) &= 0 \\ \Leftrightarrow x &= 24.\end{aligned}$$

27. Let the number be x .

$$\begin{aligned}\text{Then, } 3x^2 - 4x &= x + 50 \\ \Leftrightarrow 3x^2 - 5x - 50 &= 0 \\ \Leftrightarrow (3x + 10)(x - 5) &= 0 \\ \Leftrightarrow x &= 5.\end{aligned}$$

28. Let the number be x .

$$\begin{aligned}\text{Then, } x + \frac{1}{x} &= \frac{34}{8} \Leftrightarrow \frac{x^2 + 1}{x} = \frac{34}{8} \\ \Leftrightarrow 8x^2 - 34x + 8 &= 0 \\ \Leftrightarrow 4x^2 - 17x + 4 &= 0 \\ \Leftrightarrow (4x - 1)(x - 4) &= 0 \\ \Leftrightarrow x &= 4. \\ \left[\text{neglecting } x = \frac{1}{4}, \text{ as } x \text{ is a natural no.} \right]\end{aligned}$$

$$\therefore \text{ Required number} = 4 \times \sqrt{4} = 4 \times 2 = 8.$$

29. Let the number be x .

$$\begin{aligned}\text{Then, } \frac{2}{3}x &= \frac{25}{216} \times \frac{1}{x} \Leftrightarrow x^2 = \frac{25}{216} \times \frac{3}{2} \\ &= \frac{25}{144} \Leftrightarrow x = \sqrt{\frac{25}{144}} = \frac{5}{12}.\end{aligned}$$

30. Let the required number be x .

$$\text{Then, } x + 20 = \frac{69}{x} \Leftrightarrow x^2 + 20x - 69 = 0$$

$$\begin{aligned}\Leftrightarrow x^2 + 23x - 3x - 69 &= 0 \\ \Leftrightarrow x(x + 23) - 3(x + 23) &= 0 \\ \Leftrightarrow (x + 23)(x - 3) &= 0 \\ \Leftrightarrow x &= 3 \quad [\because x \neq -23]\end{aligned}$$

31. Let the number be x .

$$\begin{aligned}\text{Then, } x - 4 &= \frac{21}{x} \Leftrightarrow x^2 - 4x - 21 = 0 \\ \Leftrightarrow (x - 7)(x + 3) &= 0 \Leftrightarrow x = 7.\end{aligned}$$

32. Let the numbers be x and y .

$$\text{Then, } x + y = 12 \text{ and } xy = 35.$$

$$\therefore \frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy} = \frac{12}{35}.$$

33. Let the number be x .

$$\begin{aligned}\text{Then, } x + \frac{1}{x} &= 3 \left(x - \frac{1}{x} \right) \\ \Leftrightarrow \frac{x^2 + 1}{x} &= 3 \left(\frac{x^2 - 1}{x} \right) \\ \Leftrightarrow x^2 + 1 &= 3x^2 - 3 \\ \Leftrightarrow 2x^2 &= 4 \\ \Leftrightarrow x^2 &= 2 \\ \Leftrightarrow x &= \sqrt{2}.\end{aligned}$$

34. Let the numbers be a and b .

$$\text{Then, } ab = 37 \Rightarrow a = 1 \text{ and } b = 37.$$

$$\text{So, } \sqrt{b-a} = \sqrt{37-1} = \sqrt{36} = 6.$$

35. Let the numbers be a and b .

$$\text{Then, } ab = 17 \Rightarrow a = 1 \text{ and } b = 17.$$

$$\text{So, } \frac{1}{a^2} + \frac{1}{b^2} = \frac{a^2 + b^2}{a^2 b^2} = \frac{1^2 + (17)^2}{(1 \times 17)^2} = \frac{290}{289}.$$

36. Let the number be x . Then,

$$\begin{aligned}\frac{4\frac{1}{2}\left(x + 2\frac{1}{2}\right) + 3}{1\frac{1}{5}} &= 25 \Leftrightarrow \frac{\frac{9}{2}\left(x + \frac{5}{2}\right) + 3}{\frac{6}{5}} = 25 \\ \Leftrightarrow \frac{9x}{2} + \frac{45}{4} + 3 &= 25 \times \frac{6}{5} = 30 \\ \Leftrightarrow \frac{9x}{2} &= 30 - \frac{57}{4} \Leftrightarrow \frac{9x}{2} = \frac{63}{4} \\ \Leftrightarrow x &= \left(\frac{63}{4} \times \frac{2}{9}\right) = \frac{7}{2} = 3\frac{1}{2}.\end{aligned}$$

37. Let the numbers be $4x$, $5x$ and $6x$.

$$\text{Then, } \frac{4x + 5x + 6x}{3} = 25 \Leftrightarrow 5x = 25 \Leftrightarrow x = 5.$$

$$\therefore \text{ Largest number} = 6x = 30.$$

38. Let the numbers be $3x$, $4x$ and $6x$.

$$\begin{aligned}\text{Then, } 3x \times 4x \times 6x &= 1944 \\ \Leftrightarrow 72x^3 &= 1944 \Leftrightarrow x^3 = 27 \\ \Leftrightarrow x &= 3.\end{aligned}$$

$$\therefore \text{ Largest number} = 6x = 18.$$